Praxia Update

Daily orthopaedic literature digest · Saturday, 13 June 2026 · 31 new articles across 10 journals

AJSM

American Journal of Sports Medicine

June 2026 · online ahead of print · 1 new article

Partial Tenotomy and Reattachment for Chronic Proximal Hamstring Tendinopathy: Functional Outcomes, Return to Sport, and Patient Satisfaction.

Lefèvre N, Moussa MK, Valentin E, Hadjam T, Gerometta A, Grimaud O, Meyer A, Bohu Y et al.

BACKGROUND: Chronic proximal hamstring tendinopathy (PHT) can cause significant pain and functional impairment, particularly in active individuals. While nonoperative management is the first-line treatment, some patients with persistent symptoms may require surgical intervention.

PURPOSE: To evaluate the functional outcomes, return-to-sport rates, and complication profiles of patients undergoing partial tenotomy and reinsertion for chronic PHT.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This retrospective cohort study targeted patients who underwent partial tenotomy and reattachment of the proximal hamstring tendons between April 2010 and September 2022. The indication for surgery was refractory chronic PHT symptoms for >6 months. The primary outcome measure was the Parisian Hamstring Avulsion Score (PHAS). Secondary outcomes included the Tegner Activity Scale score, University of California-Los Angeles (UCLA) Activity Scale score, return to sport, postoperative patient satisfaction, and the rate of complications. In this study, the terms "partial involvement" or "complete involvement" refer to the extent of magnetic resonance imaging signal changes across the tendon, rather than to true tendon ruptures or avulsions.

RESULTS: The study included 42 patients with a mean age of 50.4 years (SD, 10.4 years). The mean follow-up was 4.1 years (range, 2.0-12.6 years […]

PubMed · DOI

CORR

Clinical Orthopaedics & Related Research

June 2026 · online ahead of print · 3 new articles

Long-term Oncologic Outcomes After Unplanned Soft Tissue Sarcoma Surgery: Implications for Surveillance Strategies.

Jahn J, Dean KK, Travis LM, Daftari M, Costello JP, Shloush M, Brutti J, Shae J et al.

BACKGROUND: Unplanned excision refers to the resection of soft tissue sarcomas without appropriate preoperative imaging, histologic confirmation, or oncologic planning. Although the short-term oncologic risks of unplanned excisions are well described, data on long-term outcomes remain limited.

QUESTIONS/PURPOSES: After controlling for tumor grade, clinical stage, Charlson comorbidity index (CCI), and surgical margin status, is unplanned excision independently associated with (1) worse overall survival, (2) local recurrence-free survival, and (3) metastatic disease-free survival compared with planned excision?

METHODS: In this study of 1100 patient records initially identified by Current Procedural Terminology code query, 403 were identified as duplicate records, 253 were excluded for insufficient baseline data or lack of cancer center entry, and 48 were excluded for insufficient follow-up or not undergoing surgery at the institution, yielding 396 patients, 279 who underwent planned excision and 117 who underwent unplanned excision. We defined an unplanned excision as resection of a soft tissue mass performed without appropriate preoperative advanced imaging (such as MRI) or histologic tissue diagnosis (such as core needle biopsy), or when imaging and biopsy findings were misinterpreted as benign, leading to resection without oncologic planning. The mean ± SD age was similar b […]

PubMed · DOI

Letter to the Editor: Editorial: AAOS Orthobiologics Registry-Sometimes, More is Less.

Wang X, Shen W

PubMed · DOI

Does Liposomal Bupivacaine Provide Superior Postoperative Analgesia Compared With Ropivacaine With Dexamethasone in Suprainguinal Fascia Iliaca Compartment Block for THA? A Randomized Controlled Trial.

Wu W, Hu X, Yang C, Sun Y, Li J, Wang L

BACKGROUND: Choosing the most effective pain management is important for enhancing functional recovery after surgery. The fascia iliaca compartment block (FICB) is recommended for analgesia after THA, but whether liposomal bupivacaine is more effective for this application remains uncertain.

QUESTIONS/PURPOSES: (1) Did a single-injection suprainguinal FICB using liposomal bupivacaine provide superior pain control by a clinically important margin? (2) What was the effect of liposomal bupivacaine on perioperative opioid consumption? (3) How did liposomal bupivacaine affect the duration of the sensory block and quadriceps motor weakness in the operated limb?

METHODS: This was a prospective, parallel-group, assessor-blinded, single-center, randomized controlled clinical trial. Patients who underwent unilateral primary THA via the posterolateral approach were included in the study. Exclusion criteria included a BMI of > 32.5 kg/m2 (which is the diagnostic criterion for moderate obesity in China), contraindications to peripheral nerve block, known allergy to study medications, psychiatric or cognitive disorders that could interfere with pain assessment, severe hepatic or renal dysfunction, and chronic pain necessitating long-term opioid therapy. According to these criteria, 76% (60 of 79) of screened patients met eligibility requirements and were subsequently randomized into either […]

PubMed · DOI

JBJS

JBJS (American)

June 2026 · online ahead of print · 1 new article

A System in Motion: The Evolution of Orthopaedic Care in Romania.

Dragosloveanu S, Nedelea DG, Scheau C

➢ Romanian orthopaedic practice has demonstrated a growing uptake of digital infrastructure, including the implementation of robotic-assisted surgical workflows and emerging applications of artificial intelligence.➢ Romania ranks among the European countries with the highest hospital bed capacity per capita, supporting the provision of sustained care for elderly patients, including those with fragility fractures and oncologic orthopaedic conditions and those undergoing complex revision surgeries.➢ The Romanian Arthroplasty Register was modernized and relaunched in 2025 as an initiative to align with other long-standing national European counterparts.

PubMed · DOI

JA

The Journal of Arthroplasty

June 2026 · online ahead of print · 8 new articles

Propensity Score-Matched Comparison of Patient-Reported Outcomes Between Crowe I Developmental Dysplasia of the Hip and Primary Osteoarthritis Following Total Hip Arthroplasty.

Saluja A, Eschert J, Ren R, Wong Z, Su E

BACKGROUND: Total hip arthroplasty (THA) is commonly performed for Crowe I developmental dysplasia of the hip (DDH) and primary osteoarthritis (OA), but comparative survivorship and patient-reported outcome measure (PROM) data remain limited with mixed findings. This study aimed to compare PROMs, revision rates, and implant survivorship between propensity-matched THA patients who had Crowe I DDH and OA.

METHODS: A retrospective review of a single surgeon's case log was performed to identify patients who had a diagnosis of DDH or OA who underwent THA. Propensity-score matching was utilized, yielding 105 DDH hips and 105 OA hips. Primary outcomes were modified Harris Hip Score (mHHS), Hip disability and Osteoarthritis Outcome Score - Joint Replacement (HOOS-JR), Visual Analog Scale (VAS) for pain, and University of California, Los Angeles Activity Scale (UCLA) collected preoperatively, at one year, and at minimum 2-year final follow-up. The secondary outcomes included implant survivorship via Kaplan-Meier analysis, multivariable logistic regression for minimal clinically important difference (MCID) and substantial clinical benefit (SCB) achievement, revision rates, and radiographic assessment of leg length discrepancy and acetabular inclination.

RESULTS: There were no significant between-group differences in any PROM at any time point (all $P > 0.05$). Analysis of covariance (ANC […]

PubMed · DOI

What Factors Increase the Risk of Dislocation after Primary Total Hip Arthroplasty Using the Direct Anterior Approach?

Murphy MP, Forti CR, Ho H, Hopper RH, Hamilton WG

BACKGROUND: While the direct anterior approach (DAA) for total hip arthroplasty (THA) is associated with a lower risk of dislocation, this complication has not been eliminated and can be acutely disabling when it occurs. This study aimed to identify patient, surgical, and radiographic factors that contribute to dislocation risk for the DAA based on primary THAs performed over a 15-year period.

METHODS: Using an institutional database, we identified 8,658 DAA THAs performed by four surgeons from March 2009 through May 2024. A Cox proportional hazards regression was performed to determine how patient demographics and implant characteristics were associated with dislocation. Patients who dislocated were also compared to a matched control cohort without dislocation to evaluate the influence of radiographic parameters, including acetabular cup orientation, offset, limb length changes, and preoperative acetabular coverage.

RESULTS: The dislocation rate among all primary THAs was 0.5% (45 of 8,656). Women had a higher dislocation rate than men (0.7 versus 0.2%), and Cox regression identified an increased risk among women with 28 mm (hazard ratio (HR) 7.5; 95% confidence interval (CI) 3.2 to 17.6) and 32 mm heads (HR 3.8; 95% CI 1.9 to 7.5). Radiographic analysis showed no significant differences in total offset (P = 0.13) or limb length change (P = 0.41), but controls demonstrated s […]

PubMed · DOI

Robotic Total Hip Arthroplasty in Atypical Hip Anatomy: Accuracy of Component Positioning and Clinical Outcomes in 192 Complex Cases.

Kayani B, Enson J, Donaldson J, Miles J, Newman S, Jayadev C, Stammers J, Skinner JA

BACKGROUND: The objectives of this study were to determine the accuracy of component positioning, patient satisfaction, functional outcomes, component survivorship, and complications of robotic total hip arthroplasty (THA) in patients who have atypical hip anatomy.

METHODS: This study included 192 robotic THAs performed in 182 patients for developmental dysplasia of the hip (n = 122), Leg-Calve-Perthes disease (n = 27), slipped capital femoral epiphysis (n = 20), previous acetabular fracture (n = 12), and skeletal dysplasia (n = 11). Predefined radiological outcomes, patient satisfaction, University of California at Los Angeles (UCLA) activity score, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), Oxford Hip Score (OHS), Forgotten Joint Score (FJS), component survivorship, and any complications were recorded. The mean follow-up time was 3.8 ± 1.7 years (range, 2.1 to 5.4).

RESULTS: Robotic THA was associated with high levels of accuracy in executing the planned horizontal (root mean square error (RMSE) 1.5 ± 1.4 mm) and vertical centers of rotation (RMSE: 1.8 ± 1.7 mm), combined offset (RMSE: 2.9 ± 3.8 mm), and leg-length correction (RMSE: 1.5 ± 1.4 mm). Acetabular component positioning within Lewinnek's safe zones was 94.7%, and Callanan's safe zones was 93.8%. The median patient satisfaction score was 90 (interquartile range (IQR), 85 to 95), the med […]

PubMed · DOI

Social Determinants of Health Z-Codes are Associated with Low Compliance with Patient-Reported Outcome Measures after Total Hip Arthroplasty.

Parry M, Ozdag Y, Kloc A, Salvato J, Mercuri JJ

INTRODUCTION: There is limited literature on compliance rates for completing patient-reported outcome measures (PROMs) after elective total hip arthroplasty (THA). Social Determinants of Health (SDOH) refers to social factors that are not classifiable elsewhere that can have major impact on healthcare outcomes. The purpose of this investigation was to assess the relationship between SDOH codes, demographic and surgical factors, and compliance rates for filling out PROMs after THA.

METHODS: This was a retrospective cohort study of elective THA cases performed between 2017 and 2022 at a single institution. Patients less than 18 years of age and patients who had less than two years of follow-up were excluded. The SDOH codes ("Z" codes), mental health codes ("F" codes), and compliance rates were recorded. The study population was binarized into two cohorts: any compliance (greater than 0%) versus no compliance (0%). Bivariate and multivariate regression (MRA) analyses were performed to assess compliance with PROMs against SDOH, demographic, and surgical factors. A total of 3,392 cases were included, of which 1,779 (52.5%) had completed at least one set of PROM questionnaires while 1,613 (47.5%) were non-compliant. Overall survey completion in the study period was 25.8%.

RESULTS: Results of the MRA indicated that having any Z-codes (odds ratio (OR) 0.355 [95% CI (confidence interv […]

PubMed · DOI

Differences Between Inpatient and Outpatient Medicare Total Knee Arthroplasty: Substantial Clinical Benefit Thresholds on Knee Injury and Osteoarthritis Outcome Score for Joint Replacement.

Patel SV, Pasqualini I, Khan ST, Elmenawi KA, Zhang C, Piuzzi NS, Cleveland Clinic Adult Reconstruction Research (CCARR) Group

BACKGROUND: The Centers for Medicare & Medicaid Services (CMS) Patient-Reported Outcome-Based Performance Measure (PRO-PM) mandates collection of patient-reported outcome measures (PROMs) after total knee arthroplasty (TKA) and defines success as achieving substantial clinical benefit (SCB) on the Knee injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS-JR) of ≥ 20 points. As TKA increasingly shifts to outpatient care and patient complexity varies by care setting, a single universal threshold may misclassify improvement. This study estimated specific KOOS-JR SCB thresholds in separate Medicare inpatient and outpatient cohorts and compared the performance of these thresholds between settings.

METHODS: A retrospective analysis of prospectively collected data on 7,926 Medicare beneficiaries ≥ 65 years who underwent primary elective TKA from 2016 to 2022 was performed. Patients were stratified by care setting into inpatient (n = 2,812) and outpatient (n = 5,114) cohorts. The SCB thresholds on KOOS-JR were derived using an anchor-based approach with the Veterans RAND 12-Item, "Compared to one year ago, how would you rate your physical health now?" as the anchor. Receiver operating characteristic analysis with Youden's index identified setting-specific cut points.

RESULTS: The KOOS-JR SCB threshold for the inpatient Medicare cohort was 29.3 points (Youden's index 0. […]

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Risk and Benefit Profile with Concurrent Nonsteroidal Anti-Inflammatory Drugs and Direct Oral Anticoagulants Following Total Knee Arthroplasty.

Paul BR, Verhey JT, Haak GM, Braithwaite CL, Tarabichi S, Christopher ZK, Spangehl MJ, Bingham JS

BACKGROUND: Nonsteroidal anti-inflammatory drugs (NSAIDs) are commonly prescribed after total knee arthroplasty (TKA) for analgesia and inflammation control in conjunction with direct oral anticoagulants (DOACs) for venous thromboembolism (VTE) prophylaxis in high-risk patients. In non-orthopaedic populations, concurrent NSAID-anticoagulant increases bleeding risk, but post-TKA safety data remain limited.

METHODS: Using a national claims database (2010 to 2023), we identified patients undergoing primary TKA who received DOAC-only VTE prophylaxis for ≥ three days. Patients co-prescribed NSAIDs were propensity score-matched 1:3 to DOAC-only patients based on demographics, comorbidities, and preoperative anticoagulant use. The matched cohort included 12,733 patients who received DOAC plus NSAID therapy and 37,127 who received DOAC alone. Outcomes included 90-day medical and surgical complications, two-year implant-related complications, and arthrofibrosis-associated outcomes. Subanalyses evaluated outcomes by NSAID type and prescription duration.

RESULTS: Combination therapy was associated with increased wound dehiscence (odds ratio (OR) 1.49, 95% confidence interval (CI) 1.20 to 1.87), surgical site infection (OR 1.71 [1.25 to 2.34]), and periprosthetic joint infection (OR 1.24 [1.01 to 1.52]), without differences in 90-day bleeding events ($P > 0.05$). At two years, NSAID plus D […]

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Complications of Ceramic-on-Ceramic Total Hip Arthroplasty with Large Heads: 10-Year Minimum Follow-Up.

Lee DH, Park JW, Jung WH, Kim TY, Ha YC, Koo KH, Lee YK

BACKGROUND: The fourth-generation ceramic bearings allow the use of larger femoral heads, but inevitably create thinner liners, raising concerns regarding liner fracture and long-term biological responses. We extended our study to evaluate late dislocation, ceramic-related complications, and long-term clinical and radiographic outcomes more than 10 years after ceramic-on-ceramic (CoC) total hip arthroplasty (THA).

METHODS: We conducted a prospective multicenter study of cementless THA using a 32/36-mm ceramic bearing. From April 2010 to February 2012, 246 patients (274 THAs) were enrolled. We evaluated ceramic fracture, noise, clinical results, radiological changes, and survival rate at a minimum of 10-year follow-up. In this study, 178 patients (100 men and 78 women, 201 hips) who had a mean age of 51 years (range, 18.9 to 77.6) at surgery were followed for a mean of 12.3 years (range, 10 to 14.6) with computed tomography (CT) scans in 47 hips (23%, 47 of 201). Kaplan-Meier survival analysis was performed with revision and reoperation as endpoints.

RESULTS: In addition to previously reported three dislocations, the new one sustained anterior instability due to lumbar degenerative kyphosis (LDK) associated with posterior impingement. Ceramic fracture did not occur in any patient. The mean Harris Hip Score was 87.0 ± 11.5 points at the latest follow-up. New focal osteolysis wa […]

PubMed · DOI

Disparities in Preoperative Weight Loss and Obesity Treatment Before Total Joint Arthroplasty.

Seward MW, Grimm JA, Bedard NA, Hannon CP, Berry DJ, Abdel MP

BACKGROUND: Institutions often implement body mass index (BMI) cutoffs for offering total hip (THA) and knee (TKA) arthroplasty. The goals of this study were to determine disparities in preoperative obesity prevalence, obesity treatment utilization, and weight loss efficacy.

METHODS: Among 21,038 primary THAs and 23,726 TKAs performed between 2002 and 2019, we identified 6,128 patients who had preoperative BMIs ≥ 30 measured one to 24 months before surgery, including 4,953 with available Area Deprivation Index (ADI) information. The mean age was 67 years (range, 19 to 97) with 55% women. The mean BMI was 36 (range, 20 to 69). The ADI was used to analyze socioeconomic status. Univariable and multivariable analyses evaluated the use of nutrition services or bariatric surgery, the likelihood of weight loss, and ADI's potential association with weight loss.

RESULTS: A few patients received preoperative weight loss medications (0.2%), nutrition services (1.6%), and/or bariatric surgery (2.6%). Compared to the best (least deprived) ADI quartile 1, the worse ADI quartiles 3 and 4 were associated with an increased odds of BMI ≥ 40 preoperatively (odds ratio (OR) 1.4, P = 0.01 and OR 1.9, P < 0.01, respectively). Women were associated with an increased odds of BMI ≥ 40 preoperatively (OR 1.6, P < 0.01) and were more likely to receive bariatric surgery (OR = 2.3, P < 0.01). Women had l […]

PubMed · DOI

JOT Journal of Orthopaedics & Traumatology

June 2026 · online ahead of print · 1 new article

Does the surgical sequence of lumbar spine fusion and total hip arthroplasty affect the dislocation and revision risk? A meta■analysis and systematic review.

Liu F, Jia C, Wu X, Zhou Y

BACKGROUND: Patients who underwent primary total hip arthroplasty (THA) after lumbar spine fusion (LSF) had a higher incidence of complications compared with those who had not previously undergone LSF. This study aimed to determine whether the timing of THA before or after LSF impacts the incidence of THA complications.

METHODS: Following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed, Web of science, Embase, and Cochrane Library from their inception until 1 October 2025. Data from all eligible published studies meeting the inclusion criteria were extracted and subsequently analyzed using a random-effects model to calculate the rates of all-cause revision, dislocation, periprosthetic joint infections, and aseptic loosening.

RESULTS: Our analysis included 10 studies involving 27,605 patients who underwent THA followed by LSF and 32,002 patients who received LSF prior to THA. There are no significant differences in the rates of dislocation (odds ratio [OR] 1.19, 95% confidence interval (CI) 0.73-1.86, p = 0.45), periprosthetic joint infections (OR 0.91, 95% CI 0.52-1.62, p = 0.76), and aseptic loosening (OR 0.89, 95% CI 0.74-1.08, p = 0.23) between the two surgical sequences, while for all-cause revision, the OR was (OR 1.01, 95% CI 0.70-1.47, p = 0.94), which reached statistical significance.

CONCLUSIONS: Compare […]

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Journal of Orthopaedic Trauma

Volume 40, Issue 7 · July 2026 · 2 new articles

Letter to the Editor Regarding "Proton Pump Inhibitors Are Associated With Increased Risk of Site-Specific Nonunion After Open Reduction Internal Fixation".

Ayas O, Acar A

PubMed · DOI

Response to "Competing-Risks Re-estimation of Five-Year Second Hip Fracture Risk: Accounting for Death as a Competing Event".

Konda SR

PubMed · DOI

Archives of Orthopaedic & Trauma Surgery

Volume 146, Issue 1 · June 2026 · 4 new articles

Does the tendon retract in flexor hallucis longus muscle laceration? A cadaveric study.

Ayd■n M, ■ahin Y, Ya■ar H, Çiçek F, Cerano■lu FG, Mert A, Ç■naro■lu S

INTRODUCTION: Flexor hallucis longus (FHL) tendon injuries are common, particularly in open injuries caused by sharp objects. However, data regarding FHL tendon retraction following hallux-level FHL tendon lacerations remain limited. This study investigated the retraction and shortening of the FHL tendon using cadaveric models.

MATERIALS AND METHODS: Six fresh-frozen below-knee amputated cadaveric feet (mean age: 65 years, range: 62-88) were utilized. FHL tendons were dissected, and lacerations were simulated at the hallux level. Tendons were subjected to various tensile forces (20-120 N) using a mechanical device. Force was measured in Newtons via a loadcell, and retraction distance was recorded in millimeters using an encoder. Data were summarized as mean ± standard deviation (SD), median, and quartiles.

RESULTS: The FHL tendon retracted gradually as applied force increased. An average retraction of 7 mm was measured at 20 N, reaching 21 mm at 60 N. Beyond 60 N (> 21 mm retraction), the other four toes also flexed. This occurs due to the anatomical connection between the FHL and flexor digitorum longus (FDL) tendons.

CONCLUSIONS: Only slight retraction was observed in hallux-level FHL tendon lacerations. At most, the findings suggest that distal FHL lacerations may exhibit limited proximal retraction under the tested conditions, which may have implications for surgical pla […]

PubMed · DOI

Early-onset rice body synovitis of the knee in children under three years: a case series and review of the literature.

Zhu L, Liu C, Liu Y, Mao B, Huang Y, Wen J, Liu Y, Qian T

OBJECTIVE: Rice body synovitis reports have been rarely reported in children under three years of age. We aimed to explore the clinical characteristics and treatment experience of rice body-like synovitis in children's knees.

METHODS: We retrospectively analyzed the clinical and imaging data of 4 cases (5 knees) of rice body-like synovitis in children treated at our hospital from January 2014 to December 2021. Among the patients, 1 was male and 3 were female, with an average age of 22.5 months. The average length of symptoms before admission was 1.86 months. Clinical presentations included swelling and limping, with no fever or rash.

RESULTS: Three patients with unilateral onset and one patient with bilateral onset underwent unilateral arthroscopic excision of synovial lesions. All 4 patients had pathological findings consistent with chronic

inflammation, leading to a postoperative diagnosis of juvenile idiopathic arthritis (JIA). Postoperatively, all patients received standardized pharmacological treatment and were followed for 24 to 48 months in collaboration with the rheumatology department. None of the 4 knees undergoing surgery experienced recurrence, and the symptoms of the knee treated conservatively resolved spontaneously. In the last follow-up session, MRI and ultrasound showed only mild synovial thickening, with effusion and loose bodies disappearing.

CONCLUSION: R […]

PubMed · DOI

Extravertebral temperature exposure during spinal radiofrequency ablation: an experimental surrogate assessment of neural injury risk.

Weigert A, Hoffmann RT, Büttner A, Zaccaria R, Wegener V, Holzapfel BM, Wegener B

BACKGROUND: Radiofrequency ablation (RFA) has become an important minimally invasive option for the treatment of painful spinal tumors and benign bone lesions. While its effectiveness within the vertebral body is well established, there is ongoing concern about unintended heat exposure of nearby neural structures. Experimental data describing how heat propagates beyond the vertebra during spinal RFA are still limited. This experimental in vitro study aims to assess the extravertebral temperature exposure during spinal radiofrequency ablation as a surrogate parameter for potential neural injury risk under controlled conditions.

METHODS: This experimental in vitro study was performed on four fresh-frozen human lumbar spines. Radiofrequency ablation was carried out using a commercially available system and standard clinical protocols. Two anatomical configurations were examined: ablation within the vertebral body and ablation within the pedicle. Temperatures were measured at predefined locations outside the vertebra, including the spinal canal and cortical boundaries, using multiple thermocouples. To aid interpretation of radiofrequency-related measurement artifacts, an additional control setup using an electrothermal heating system without radiofrequency energy was included. Temperature data were evaluated descriptively.

RESULTS: In the control experiments, temperature profiles […]

PubMed · DOI

Giant cell tumor of bone: exploring HtrA1 as a potential predictor of recurrence.

Mirioglu A, Dalkir KA, Gokmen MY, Bagir M, Ates KE, Deveci MA, Gönlü■en G, Ozbarlas HS

INTRODUCTION: Giant cell tumor of bone (GCTB) is a locally aggressive tumor with a considerable risk of recurrence, but reliable predictive markers are limited. High-temperature requirement A serine peptidase 1 (HtrA1) has been implicated in the progression of several cancers and may also play a role in GCTB recurrence.

MATERIALS AND METHODS: This retrospective study included 34 patients with GCTB treated with curettage between 2002 and 2016. HtrA1 expression was evaluated separately in giant cells (GC) and mononuclear cells (MNC) using immunohistochemistry. Based on high or low expression in each cell type, patients were categorized into four expression patterns: High GC / High MNC, High GC / Low MNC, Low GC / High MNC, and Low GC / Low MNC. Clinical data, including recurrence status and time to recurrence, were analyzed.

RESULTS: The High GC / Low MNC group showed the highest recurrence rate at 71.4%, compared with 28.6% in the High GC / High MNC group and 31.6% in the Low GC / Low MNC group. No recurrence was observed in the Low GC / High MNC group, which included only one patient. The High GC / Low MNC group showed a non-significant trend toward higher recurrence compared with all other groups combined, with a similar trend toward shorter recurrence-free survival.

CONCLUSION: HtrA1 expression patterns in GCTB may provide preliminary insight into recurrence risk. Although […]

PubMed · DOI

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Injury

Volume 57, Issue 8 · June 2026 · 8 new articles

More than meets the eye: the continuing importance of tertiary trauma survey even in moderate (ISS 9-15) trauma.

Knight C, Burke JR, Hind S, Smith SR

INTRODUCTION: Tertiary Trauma Surveys (TTS) are designed to identify injuries which are not recognised during the primary and secondary surveys. These injuries occur in up to 40% of patients and are commonly referred to in the literature as "missed", we prefer the term "unrecognised" as this term does not infer the failure of primary and secondary survey, rapid assessments designed to identify life and limb threatening injuries. The incidence and definition of unrecognised injuries varies across studies, and several risk factors have been identified internationally. The aim of our study was to investigate the sensitivity and specificity of TTS and the incidence and risk factors for unrecognised injuries in a large UK Major Trauma Centre (MTC).

METHODS: Data was prospectively collected for all major trauma patients admitted to our MTC over six months. All patients over 16 years with an ISS≥ 9 were included. Patients with a length of stay of less than 24 h or palliated were excluded. All patients were examined using a specific TTS protocol. Patient records were reviewed one year post admission to identify any injuries diagnosed after discharge.

RESULTS: A total of 293 patients were included. Unrecognised injuries were identified at TTS in 18.8% (55/293) patients, most commonly affecting the face or head and eyes. Patients with unrecognised injuries had a higher admission ISS (p […]

PubMed · DOI

CTA-detected incidence and descriptive fracture patterns of arterial injury in 539 surgically treated tibial plateau fractures.

Valderrama J, Narváez F, Cancino J, Raffo JF, Bosselin R, Zamora F, Siegel S, Scheu M et al.

BACKGROUND: Tibial plateau fractures (TPF) are complex injuries associated with significant soft tissue and periarticular complications. Although vascular injuries are well described in lower limb trauma, their incidence in tibial plateau fractures has not been clearly established.

METHODS: Retrospective cohort study including patients with surgically treated TPF at a level I trauma center between 2016 and 2024. CTA was selectively performed based on clinical judgment. Post hoc descriptive risk scenarios were constructed to organize observed fracture patterns. Analysis was primarily descriptive.

RESULTS: 539 surgically treated TPF were included. Mean age was 43.7 years (18-75), 73.84% were male. Left knee was affected in 54.92%. High-energy trauma accounted for 70.87%. The CTA yield was 3.66% (6/164), corresponding to an overall observed incidence of 1.11% (6/539). Certain fracture patterns and clinical scenarios were more frequently observed among cases with arterial injury; however, these observations are based on very few events and should be interpreted cautiously as descriptive findings only.

CONCLUSION: The CTA-detected incidence of arterial injury in this series of surgically treated tibial plateau fractures was 1.11%. Certain fracture patterns and clinical contexts were observed among the small number of cases with arterial injury; however, these findings are descrip […]

PubMed · DOI

Diagnostic value of serum troponin in stable patients at risk for blunt cardiac injury.

Cleary HL, Bernard MD, Davenport DL, Bernard AC

INTRODUCTION: There are no gold standard criteria for diagnosing blunt cardiac injury (BCI). While the combination of electrocardiogram (ECG) and serum troponin is considered the most sensitive screening method for those being considered for discharge, little evidence exists regarding the diagnostic value and clinical utility of initial and serial serum troponin levels in stable patients who are being admitted with a possible BCI. Therefore, this study seeks to determine if troponin level is an independent predictor of adverse cardiac events in stable,

admitted patients at risk for BCI.

METHODS: This was a five-year retrospective study using the trauma database at a University Level I Trauma Center. The study population included adult trauma patients presenting with a physician diagnosis of BCI or a sternal fracture who met prespecified stability criteria (SBP ≥ 90 mmHg, HR < 110bpm, shock index < 1, GCS ≥ 14). The data collected included all troponin values and ECG interpretations as well as echocardiogram interpretations, when performed. A patient was classified as having an adverse cardiac event if they were diagnosed with a new arrhythmia requiring treatment, had cardiac surgery, or suffered cardiac-related mortality. Sensitivity, specificity, positive and negative likelihood ratios, and diagnostic odds ratios were calculated to analyze the diagnostic performance of the di […]

PubMed · DOI

Risk Factors for Failure of Repair of Diaphyseal Tibial Nonunions: A Retrospective Cohort Study of 108 Patients.

Fan S, Wagner RK, Lehle CH, Musick AN, Gregg AT, Janssen SJ, Wijffels MME, Stenquist DS et al.

OBJECTIVES: To determine rates and risk factors for additional nonunion surgery and final radiographic healing following diaphyseal tibial nonunion repair.

METHODS: All consecutive patients aged ≥18 years who underwent diaphyseal (AO/OTA 42) tibial nonunion repair with (exchange) intramedullary nailing (IMN) or open reduction and internal fixation (ORIF) between 2001 and 2024 with ≥6 months follow-up were included. The primary outcome was the rate of additional nonunion surgery. The secondary outcome was radiographic healing (defined as Modified Radiographic Union Score for Tibia [mRUST] ≥11) at final follow-up. Multivariable logistic regression was used to determine the association of patient, nonunion, and treatment characteristics with additional nonunion surgery.

RESULTS: A total of 108 patients were included (70% male, median age 42 years [IQR 28-53]). Sixty patients (56%) were treated with IMN, of whom 54 (50%) underwent exchange IMN, and 48 (44%) were treated with ORIF. Autograft was used in 39 patients (36%) and bone graft substitutes in 21 (19%). Twenty-seven patients (25%) required additional nonunion surgery, of whom 5 (4.6%) ultimately underwent amputation. In multivariable analysis, current smoking (OR: 4.28, $p = 0.015$), initial open fracture (OR: 3.94, $p = 0.045$), and presence of a bone defect (OR: 9.20, $p < 0.001$) were associated with increased odds of addition […]

PubMed · DOI

Superior survival, hemodynamics, and metabolic recovery using PEG-20k-based low volume resuscitation in a polytrauma hemorrhagic shock model in rats.

Khoraki J, Payne C, Lust D, Mangino MJ, Liebrecht LK

BACKGROUND: Hemorrhagic shock is a major cause of preventable civilian and military trauma death. Although whole blood and blood products remain the standard of care, field■stable alternatives are needed for resource■limited environments and supply■chain disruptions. Blood■based resuscitation may also fail to optimally restore microvascular perfusion despite improved macrohemodynamics, contributing to ongoing ischemia-reperfusion injury, dysfunctional molecular cell dynamics, and capillary cell swelling. Prior controlled hemorrhage studies in rats and swine demonstrated superior survival, mean arterial pressure (MAP), and lactate clearance with polyethylene glycol 20,000 (PEG■20k) compared with crystalloid, colloid, and whole blood controls. These beneficial effects are attributed to the capacity of PEG-20k to augment water transfer from swollen ischemic cells into the vascular space, thereby decompressing capillary beds and restoring effective tissue perfusion. This study evaluated PEG■20k in a severe polytrauma rat model with combined controlled and uncontrolled hemorrhage.

METHODS: Adult male Sprague-Dawley rats (250-400-g) underwent a midline laparotomy, 75% tail amputation, and dual splenic transections. The spleen bled freely into the closed abdomen (uncontrolled, non-compressible hemorrhage), while the tail was intermittently clamped (controlled, compressible hemorrhage […]

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The forequarter lateral implosion injury: A diagnostic paradigm and Ch-Sh (ch■-sh■) classification.

George A, Paull TZ, Winter JD, Levine J, Cole PA

BACKGROUND: The lateral impaction force against the shoulder often causes multiple lesions to the forequarter including the chest wall. The purpose of this study is to describe the injury patterns and associated pathology in patients with combined chest wall and shoulder girdle injuries- forequarter lateral implosion injury and organize the findings into a classification framework to predict morbidity.

METHODS: Three hundred fifty-four consecutive patients with two or more rib fractures and ipsilateral shoulder girdle injury from a Level 1 trauma center, between 2018-2022, were retrospectively reviewed. Ch-Sh is pronounced ch■ sh■. Chest injuries were grouped into three categories: Ch1: 2-4 fractured ribs, Ch2: 5 + fractured ribs, or Ch3: 2 + fractured ribs with contralateral rib fractures and/or sternal fracture. Shoulder girdle injuries were grouped into: Sh1: clavicle or acromion fracture, and/or AC/SC joint dislocation, Sh2: scapula and/or proximal humerus fracture, or Sh3: combination of Sh1 and Sh2 injuries (clavicular chain and scapula/humerus). There were 9 total subgroups. Demographics, injury characteristics, associated injuries, and outcomes were recorded. Inter and intraobserver reliability of the classification system was studied.

RESULTS: As chest wall injuries scaled from Ch1 to Ch3, there were significantly higher rates of ICU admission, pneumothorax, pulmonar […]

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The fate of stage I Masquelet cement spacers in long bone reconstruction for acute fractures.

Dekle JK, Sisk CK, Litten RM, Alcaide DM, McIlwain RN, Wilson AL, Manson HC, Spitler CA et al.

INTRODUCTION: The Masquelet technique is widely used for reconstruction of segmental long bone defects following acute fractures; however, not all patients progress to the second-stage bone grafting procedure. The purpose of this study was to characterize Stage II completion following Stage I Masquelet cement spacer placement for acute long bone fractures, compare characteristics and outcomes between patients who underwent Stage I only versus completed Stage II reconstruction, and identify factors associated with progression to Stage II bone grafting.

METHODS: A retrospective review was conducted of adult patients who underwent Masquelet reconstruction for acute long bone fractures at a Level I trauma center between 2014 and 2024. Patients were stratified by completion of Stage II reconstruction. Demographic, injury-related, surgical, and clinical outcome variables were collected. Multivariable logistic regression was performed to identify factors independently associated with Stage II completion.

RESULTS: A total of 156 patients met inclusion criteria, of which 48 (30.8%) underwent Stage I only and 108 (69.2%) completed both stages. Patients who remained at Stage I underwent fewer orthopaedic operations (3.2 vs. 4.2, $p = 0.001$) and initial bony defect size did not differ between groups (84.1 vs. 78.8 mm, $p = 0.423$). Among patients with at least 6 months of follow-up, rates o […]

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Rethinking the Lisfranc injury: Rationale for and development of a new classification system for Lisfranc injuries.

Cawley WCF, Hwang P, Mittal R, Zochowski Y, Darwish I, O'Keefe R, Grogin J, Sivakumar B et al.

BACKGROUND: The Lisfranc joint comprises the articulations between the cuneiforms and bases of the metatarsals and is susceptible to a variety of complex injuries which are frequently missed or diagnosed late. This can result in morbidity with poor outcomes, and renders it challenging to standardise management and allow prognostication. Existing classification systems, including the Myerson-Hardcastle (MHC) and Nunley & Vertullo classifications, do not consider the propagation of force through the midfoot, which is key to identifying structures which may require stabilization. They also demonstrate poor clinical utility due to poor correlation with outcomes. This study aimed to develop a new classification system for Lisfranc injuries using data from a large single-surgeon case series from a tertiary level trauma hospital in Australia. The primary outcome measures for this study were inter- and intra-observer reliability.

METHODS: The Suthersan classification was developed to address shortfalls in current classification systems. Patient records were manually searched for Lisfranc injuries. Direct 'crush' Lisfranc injuries were excluded due to unpredictable injury patterns. Indirect Lisfranc injuries were included in analysis. Two orthopaedics foot and ankle surgeons and one hand surgeon reviewed patient radiographs and classified them using the MHC and Suthersan classification […]

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Denis pattern-dependent biomechanical behavior of posterior trans-iliac plate fixation compared with bilateral triangular osteosynthesis in unilateral sacral fractures: a finite element study.

Yang J, Böker KO, Li C, Schilling AF, Ritter Z, Acharya M, Lehmann W

OBJECTIVE: Posterior trans-iliac plate osteosynthesis (TPO) is commonly used for the stabilization of unilateral vertically unstable sacral fractures. However, its biomechanical performance across different Denis fracture patterns remains unclear. This study aimed to investigate the fracture pattern-dependent biomechanical behavior of posterior trans-iliac plate osteosynthesis (TPO) in unilateral vertically unstable sacral fractures, using bilateral triangular osteosynthesis (BTO) as a high-stability reference construct.

METHODS: A validated three-dimensional finite element model of the lumbopelvic complex was developed to simulate unilateral vertically unstable sacral fractures corresponding to Denis type I, II, and III patterns. Two posterior fixation strategies, TPO and BTO, were constructed for each fracture type. Seven physiological loading conditions were applied, including standing, flexion, extension, axial rotation, and lateral bending. Von Mises stress distribution and fracture displacement were evaluated. Fracture micromotion was quantified by calculating relative displacement changes between paired points along the fracture gap.

RESULTS: When interpreted relative to the high-stability BTO reference construct, TPO showed fracture pattern-dependent variation in fracture displacement. The difference was most pronounced in Denis type I fractures and progressively decr […]

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Factors associated with in-hospital mortality in necrotising soft tissue infections. a multicentre retrospective cohort study.

Podda M, Ceresoli M, Virdis F, Rosa F, Pilia T, Murzi V, Dessì A, Tedesco S et al.

PURPOSE: Necrotising soft tissue infections (NSTIs) are rare but life-threatening conditions associated with high mortality rates. This multicentre study aimed to identify admission variables associated with in-hospital mortality.

METHODS: This retrospective study included adult patients with surgically confirmed NSTIs treated at four high-volume academic referral centres in Italy between 2010 and 2024. Demographic, clinical, physiological, and laboratory variables available at hospital admission were analysed. Categorical variables were compared between survivors and non-survivors using the chi-square test or Fisher's exact test, while quantitative variables were compared using the Student's t test or Mann-Whitney U test. Multivariable logistic regression analyses were performed to identify independent predictors of in-hospital mortality. Results were reported as ORs with 95%CI. A prognostic nomogram was developed from the final multivariable model, including variables independently associated with mortality.

RESULTS: A total of 379 patients were included. In-hospital mortality was 16.7%. In subgroup comparisons, mortality was 17.0% in necrotising fasciitis and 22.4% in Fournier's gangrene ($P = 0.275$). In the overall NSTI population, age (aOR 1.061, 95%CI 1.037-1.087) and serum creatinine at admission (aOR 1.301, 95%CI 1.040-1.629) were independently associated with mortalit […]

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Impact of Preoperative Osteoporotic Medications on Total Hip Arthroplasty Outcomes.

Antonioli SS, Ruff G, Leung N, Patel A, Schwarzkopf R, Cohen-Rosenblum A

BACKGROUND: Osteoporosis (OP) is a common comorbidity in patients undergoing total hip arthroplasty (THA) and is a known risk factor for poor postoperative outcomes such as periprosthetic fracture (PPF). The impact of preoperative OP medications in patients with OP undergoing THA remains unclear. This study aimed to compare THA outcomes by OP diagnosis and preoperative bone strengthening medication usage.

METHODS: This was a retrospective review of primary elective THAs from June 2011-January 2024. Patients were stratified by OP diagnosis and OP medication usage, then propensity matched in a 1:2:3 ratio by age, sex, body mass index, and comorbidities. The resulting cohorts: (1) OP and medication usage for at least 1 year preoperatively and within 7 years of the procedure (n = 296), (2) OP and no medication usage (n = 592), and (3) no diagnosis of OP and no medication usage (n = 888) were then compared for postoperative outcomes.

RESULTS: Patients in cohort 1 were more likely to visit the emergency department ($P = .043$) or be readmitted for a surgical complication ($P = .007$) within 90 days. They were more likely to undergo posterior approach than cohorts 2 and 3 ($P = .016$), had the highest occurrence of cemented femoral stems (38.2%, $P < .001$), and had a higher incidence of revision for PPF (3.4%, vs 0.8% [cohort 2] and 1.2% [cohort 3], $P = .009$). Of the 12 revisions due to pe […]

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